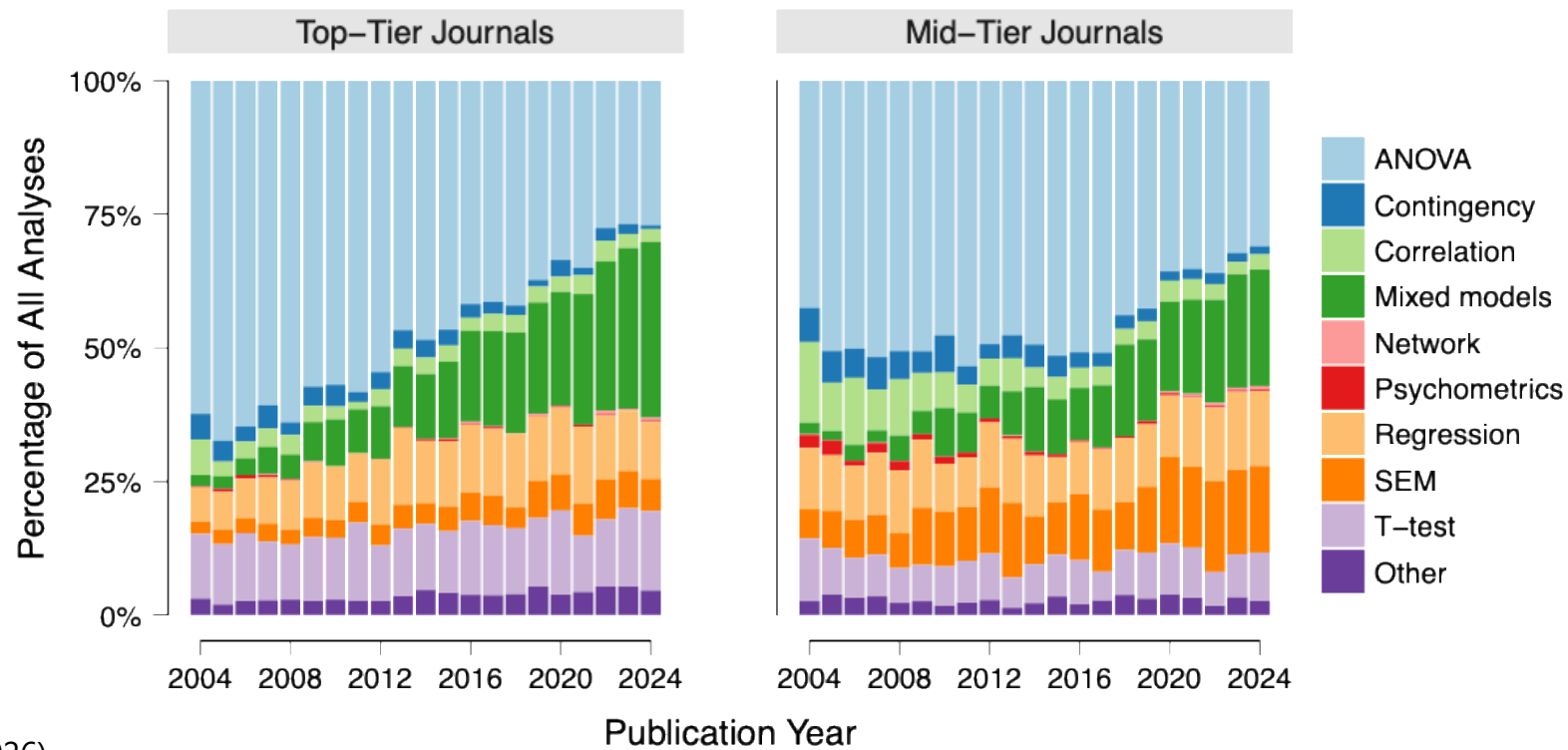


# Contents – Day 2

1. Bayesian multi-model inference
- 2. Mixed-effects regression in R with *brms***
3. How to conduct and report a Bayesian analysis
4. Bring your own data
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# Mixed effects regression – Why?



Pfadt et al. (2026)

# Mixed effects regression – Bayesian benefits

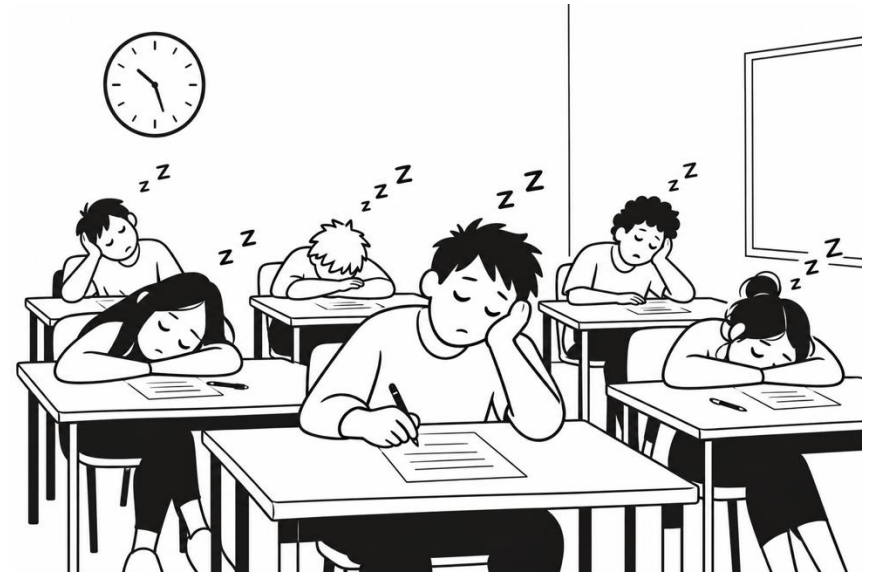
- Boundary estimation behaves:
  - Maximum likelihood estimates a single value, that sometimes ends up at the boundary where uncertainty is impossible
  - The posterior distribution can concentrate near the boundary without collapsing on it
- Propagated uncertainty:
  - Maximum likelihood estimates the variance components first, then treats them as known when computing everything else, their uncertainty gets lost when estimating parameters that depend on the variance
  - The Bayesian model estimates everything jointly, variance uncertainty propagates and all estimates have a posterior distribution
- All the other benefits: posterior distributions, credible intervals, evidence for hypotheses



# Mixed effects regression – Theory I

Artificial example:

- Large investigation into how sleep affects test scores of high school students across all the schools in a city
- $n = 350$  students in 10 schools, outcome *test score*, predictor *sleep hours*
- Naïve approaches:
  - Treat all 350 test scores as independent and fit a linear regression:  $Score = \beta_0 + \beta_1 Sleep$ , overstating evidence, biased (small) standard errors
  - Average each school's scores: Interesting variation is lost
  - Fit separate regression for each school



# Mixed effects regression – Theory II

Multilevel model with two levels:

- Level 1 – within a school ( $i$  indexes schools,  $j$  indexes students within that school):

$$Score_{ij} = \beta_{0i} + \beta_{1i}Sleep_{ij} + \epsilon_{ij}, \quad \epsilon_{ij} \sim \text{Normal}(0, \sigma^2)$$

- Level 2 – between schools:

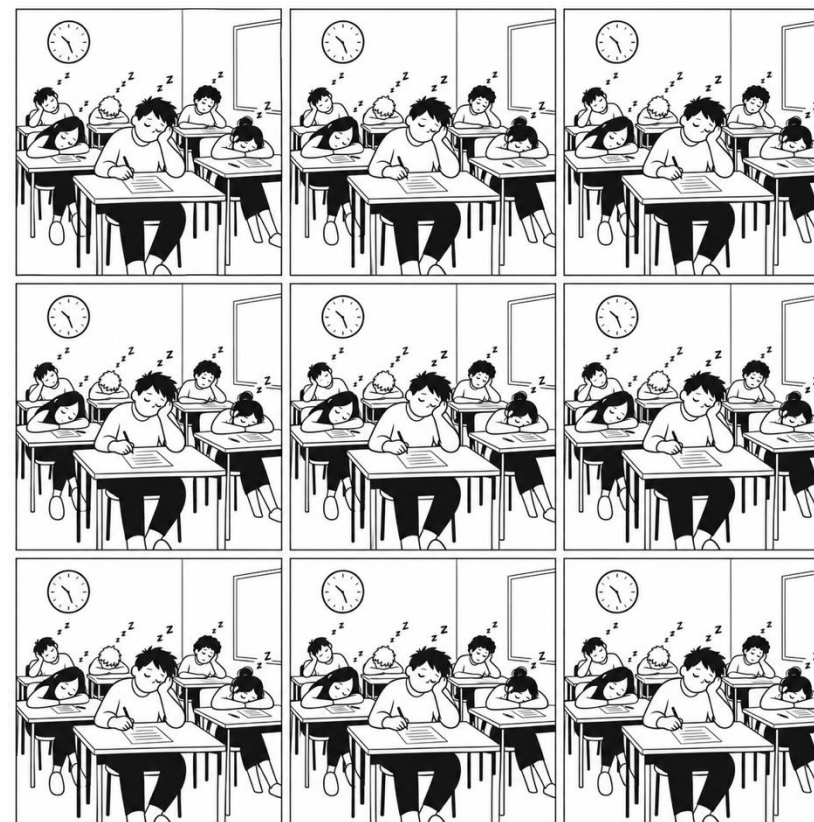
$$\beta_{0i} = \gamma_{00} + u_{0i}$$

$$\beta_{1i} = \gamma_{10} + u_{1i}$$

- The school specific parameters  $\beta_{0i}$  and  $\beta_{1i}$  are called random intercept and random slope
- Random intercept and random slope consist of a grand mean and each school's offset
- Each school's random effect has a multivariate distribution:

$$\begin{pmatrix} u_{0i} \\ u_{1i} \end{pmatrix} \sim MVN \left( 0, \begin{pmatrix} \sigma_{u0}^2 & \sigma_{u01} \\ \sigma_{u01} & \sigma_{u1}^2 \end{pmatrix} \right)$$

- $\sigma_{u0}^2$ : variance of random intercept: How much schools differ in their baseline
- $\sigma_{u1}^2$ : variance of random slope: How much schools differ in their slope
- $\sigma_{u01}$ : covariance between the intercepts and slopes



# Mixed effects regression – Bayesian analysis

## Priors:

- Fixed effects ( $\gamma$ ): Normal, Student-t, *brms* default is a flat prior (uniform over real line)
- Between group SDs ( $\sigma_u$ ): Half-Normal, half-Student-t, Exponential, *brms* default on the SD half-Student-t(3)
- Correlation ( $\rho_u$  from  $\sigma_{u01}$ ): LKJ, *brms* default for one correlation is LKJ(1)
- Residual SD ( $\sigma$ ): same as between group SDs

## Likelihood:

$$p(\text{Score}_{ij} \mid \beta_{0i}, \beta_{1i}, \sigma) = \text{Normal}(\beta_{0i} + \beta_{1i}\text{Sleep}_{ij}, \sigma)$$

- The likelihood looks simple, but the betas carry a lot of weight
- It is not analytically tractable
- -> MCMC

# Mixed effects regression - MCMC

Markov Chain Monte Carlo:

- Andrey Markov: next step depends only on current position
- Monte Carlo: Random repeated draws (game like)
- Finding the posterior by repeated random draws where the sampling process is a Markov chain
- Drawing a large sequence of samples from the posterior without ever needing the formula for the posterior itself

- How? 🤔 ➡

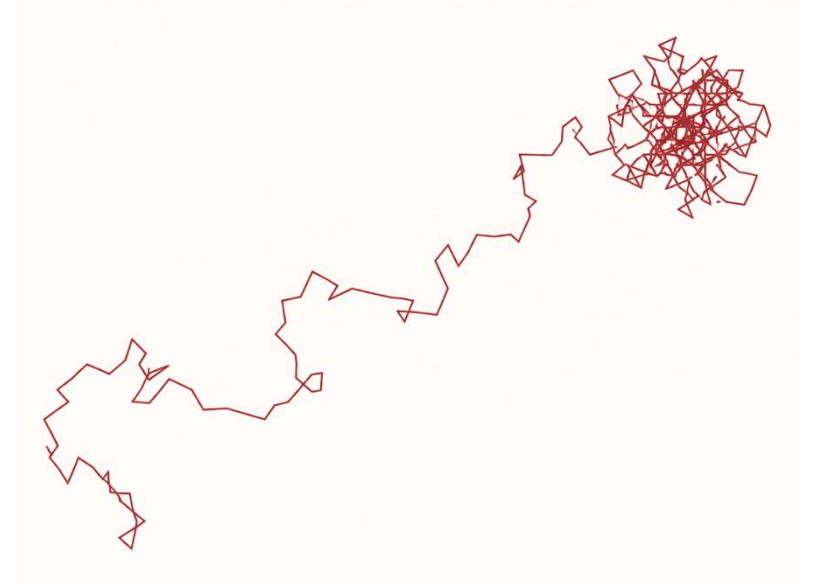
- The mechanism:

We start at some parameter value, plug it into likelihood  $\times$  prior. We propose a nearby value, plug that in too. If the new number is bigger, we move there. If it's smaller, we still might move, if it's much smaller we do not move. The explored space is the posterior.



# Mixed effects regression – MCMC sampling

- We run the sampling algorithm for many *iterations* (thousands)
- We start the sampling algorithm multiple times (*chains*) with different starting values
- We discard the first few (hundreds) of samples (*burnin*)
- When the sampled values represent the target distribution we say it *converged*
- If we constructed priors and likelihood correctly, the target distribution represents the posterior
- We have to check convergence!

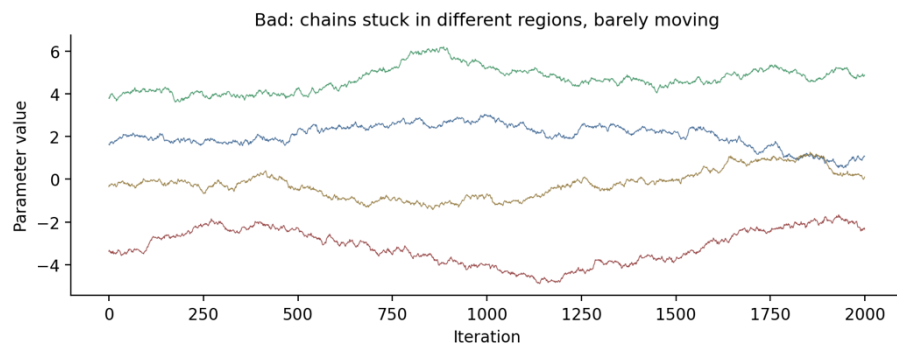
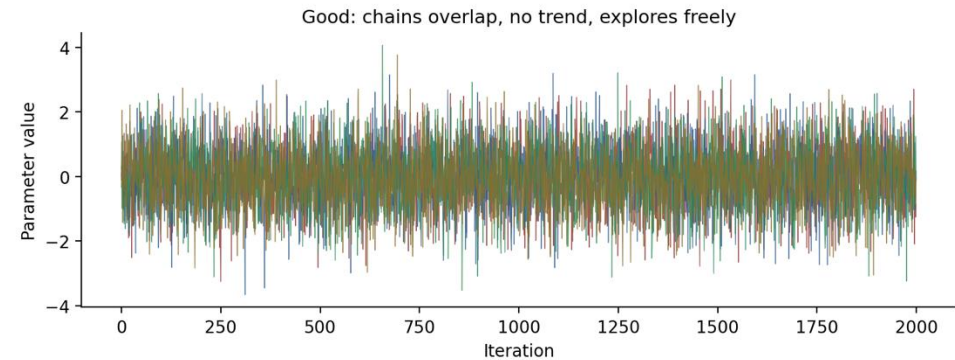
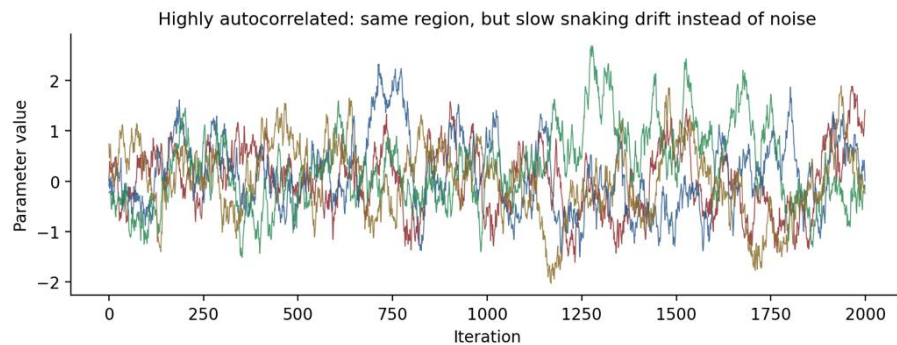




# Mixed effects regression – MCMC convergence

*Convergence* can be assessed with metrics:

- R-hat: compares within chains variance with between chain variance
- ESS: Effective sample size, measures the autocorrelation in samples, the less correlation the better and the higher ESS
- Traceplots:



# Mixed effects regression – *brms*



- An R package for fitting **B**ayesian **r**egression **m**odels using **S**tan
- Linear and generalized linear models, multilevel models, non-linear and distributional models
- A wide range of response distributions: Gaussian, binomial, Poisson, ordinal, and many more
- Missing data, measurement error, censored and truncated outcomes
- Prior controls
- Posterior checks and diagnostics

# Mixed effects regression – *brms*



$$Score_{ij} = \beta_{0i} + \beta_{1i}Sleep_{ij} + \epsilon_{ij},$$

$$\beta_{0i} = \gamma_{00} + u_{0i} \text{ (random intercept)}$$

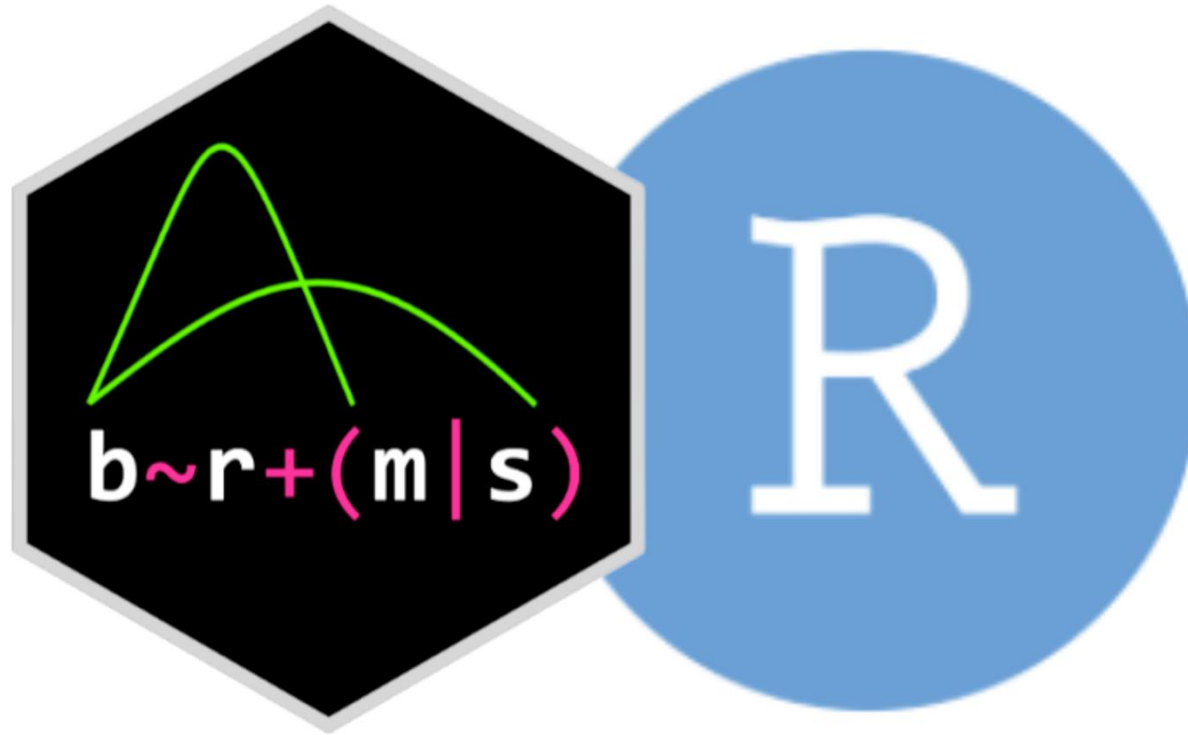
$$\beta_{1i} = \gamma_{10} + u_{1i} \text{ (random slope)}$$

- Formula syntax in R is very close to `lm()` and `lme4::lmer()`
- The intercept is by default included

```
outcome ~ predictors + (what_varies_by_group | grouping_variable)
```

```
brm(Score ~ Sleep + (1 + Sleep | school), data = dat)
```

# Sleep-School example in *brms*



-> But first some more stuff about priors

# Mixed effects regression – Food example

Example data:

- 79 participants, each rated 24 foods on tastiness, 1896 ratings
- Question: Does the cultural association (Southernness) of a food predict how tasty people find it
- Level 1 – within a person ( $i$  indexes a person;  $j$  indexes the food item):

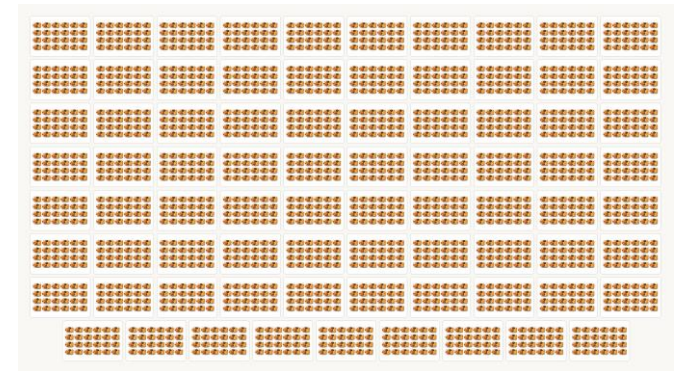
$$Tasty_{ij} = \beta_{0i} + \beta_{1i}Southernness_{ij} + \epsilon_{ij}$$

- Level 2 – between people, but we bring in the predictor Southern identification:

$$\beta_{0i} = \gamma_{00} + \gamma_{01}SouthernID_i + u_{0i}$$

$$\beta_{1i} = \gamma_{10} + \gamma_{11}SouthernID_i + u_{1i}$$

- $\gamma_{11}$  is the interaction of *Southernness* and *SouthernID*: How much a person's *Southernness* slope changes with how strongly they identify as Southern



# Mixed effects regression – *brms*



$$Tasty_{ij} = \beta_{0i} + \beta_{1i}Southernness_{ij} + \epsilon_{ij}$$

$$\beta_{0i} = \gamma_{00} + \gamma_{01}SouthernID_i + u_{0i}$$

$$\beta_{1i} = \gamma_{10} + \gamma_{11}SouthernID_i + u_{1i}$$

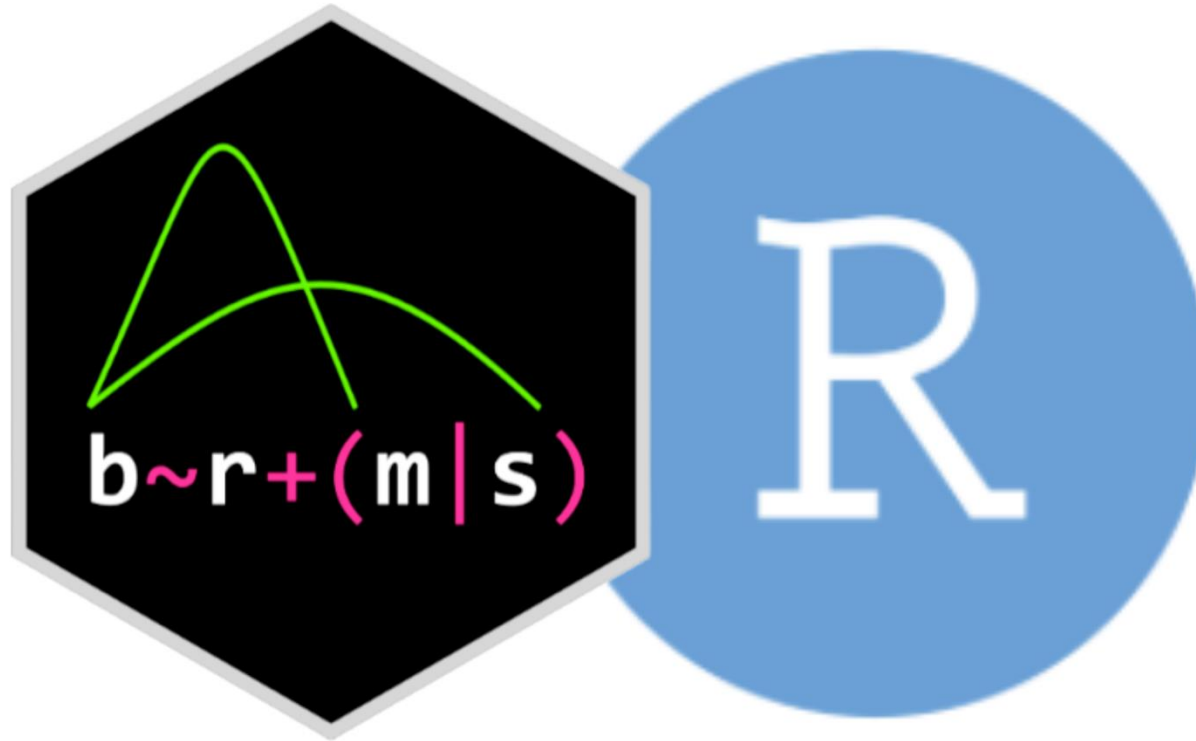
- Formula syntax in R is very close to `lm()` and `lme4::lmer()`:

```
outcome ~ predictors + (what_varies_by_group | grouping_variable)
```

```
Tasty ~ Southernness * SouthernID + (1 + Southernness | participant)
```

```
Tasty ~ Southernness + SouthernID + Southernness:SouthernID + (1 + Southernness | participant)
```

# Open food\_ratings\_brms\_exercises.R



# Contents – Day 2

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# How to conduct and report a Bayesian analysis – Planning

1. Specify the goal of the analysis and sampling plan:

- Parameter estimation
- Hypothesis testing (one-sided, two-sided)
- Both
- How you intend to collect the data

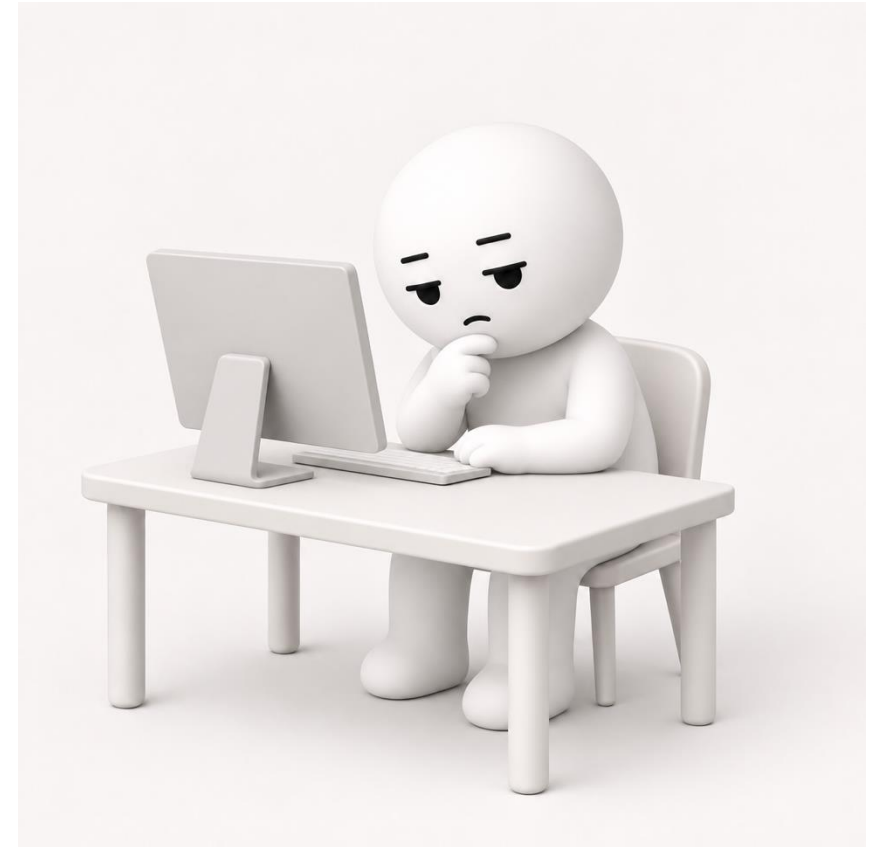
2. Define the statistical model:

- Structure, parameters, likelihood, priors
- Multiverse analysis?

3. Report and justify the prior:

- Prior distributions, prior hyperparameters
- Prior predictive checks

4. Preregistration



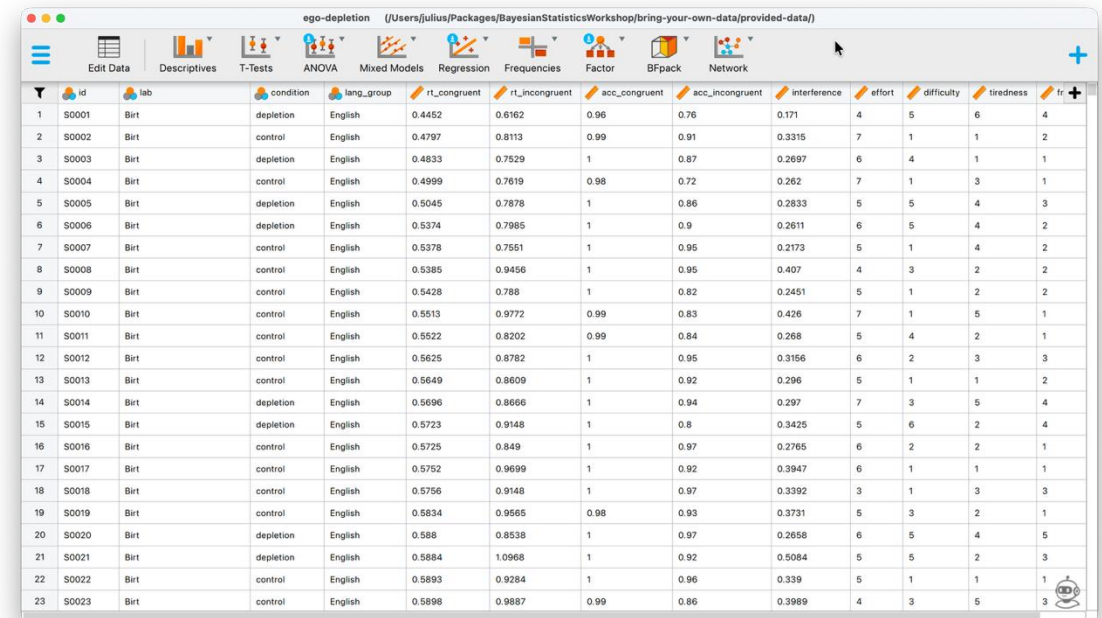
# How to conduct and report a Bayesian analysis – Executing

## 5. Assumption checks

- Normality, outliers, functional form, etc.

## 6. Running the analysis:

- MCMC: Adjust sampling to your needs (more iterations, more chains, or a finer step size), aim for convergence
- Posterior predictive checks (model fit)
- Possibly: Principled model comparison
- Prior sensitivity analysis



ego-depletion (/Users/julius/Packages/BayesianStatisticsWorkshop/bring-your-own-data/provided-data/)

|    | id    | lab  | condition | lang_group | rt_congruent | rt_incongruent | acc_congruent | acc_incongruent | interference | effort | difficulty | tiredness | tr |
|----|-------|------|-----------|------------|--------------|----------------|---------------|-----------------|--------------|--------|------------|-----------|----|
| 1  | S0001 | Birt | depletion | English    | 0.4452       | 0.6162         | 0.96          | 0.76            | 0.171        | 4      | 5          | 6         | 4  |
| 2  | S0002 | Birt | control   | English    | 0.4797       | 0.8113         | 0.99          | 0.91            | 0.3315       | 7      | 1          | 1         | 2  |
| 3  | S0003 | Birt | depletion | English    | 0.4833       | 0.7529         | 1             | 0.87            | 0.2697       | 6      | 4          | 1         | 1  |
| 4  | S0004 | Birt | control   | English    | 0.4999       | 0.7619         | 0.98          | 0.72            | 0.262        | 7      | 1          | 3         | 1  |
| 5  | S0005 | Birt | depletion | English    | 0.5045       | 0.7878         | 1             | 0.86            | 0.2833       | 5      | 5          | 4         | 3  |
| 6  | S0006 | Birt | depletion | English    | 0.5374       | 0.7985         | 1             | 0.9             | 0.2611       | 6      | 5          | 4         | 2  |
| 7  | S0007 | Birt | control   | English    | 0.5378       | 0.7551         | 1             | 0.95            | 0.2173       | 5      | 1          | 4         | 2  |
| 8  | S0008 | Birt | control   | English    | 0.5385       | 0.9456         | 1             | 0.95            | 0.407        | 4      | 3          | 2         | 2  |
| 9  | S0009 | Birt | control   | English    | 0.5428       | 0.788          | 1             | 0.82            | 0.2451       | 5      | 1          | 2         | 2  |
| 10 | S0010 | Birt | control   | English    | 0.5513       | 0.9772         | 0.99          | 0.83            | 0.426        | 7      | 1          | 5         | 1  |
| 11 | S0011 | Birt | control   | English    | 0.5522       | 0.8202         | 0.99          | 0.84            | 0.268        | 5      | 4          | 2         | 1  |
| 12 | S0012 | Birt | control   | English    | 0.5625       | 0.8782         | 1             | 0.95            | 0.3156       | 6      | 2          | 3         | 3  |
| 13 | S0013 | Birt | control   | English    | 0.5649       | 0.8609         | 1             | 0.92            | 0.296        | 5      | 1          | 1         | 2  |
| 14 | S0014 | Birt | depletion | English    | 0.5696       | 0.8666         | 1             | 0.94            | 0.297        | 7      | 3          | 5         | 4  |
| 15 | S0015 | Birt | depletion | English    | 0.5723       | 0.9148         | 1             | 0.8             | 0.3425       | 5      | 6          | 2         | 4  |
| 16 | S0016 | Birt | control   | English    | 0.5725       | 0.849          | 1             | 0.97            | 0.2765       | 6      | 2          | 2         | 1  |
| 17 | S0017 | Birt | control   | English    | 0.5752       | 0.9699         | 1             | 0.92            | 0.3947       | 6      | 1          | 1         | 1  |
| 18 | S0018 | Birt | control   | English    | 0.5756       | 0.9148         | 1             | 0.97            | 0.3392       | 3      | 1          | 3         | 3  |
| 19 | S0019 | Birt | control   | English    | 0.5834       | 0.9565         | 0.98          | 0.93            | 0.3731       | 5      | 3          | 2         | 1  |
| 20 | S0020 | Birt | depletion | English    | 0.588        | 0.8538         | 1             | 0.97            | 0.2658       | 6      | 5          | 4         | 5  |
| 21 | S0021 | Birt | depletion | English    | 0.5884       | 1.0968         | 1             | 0.92            | 0.5084       | 5      | 5          | 2         | 3  |
| 22 | S0022 | Birt | control   | English    | 0.5893       | 0.9284         | 1             | 0.96            | 0.339        | 5      | 1          | 1         | 1  |
| 23 | S0023 | Birt | control   | English    | 0.5898       | 0.9887         | 0.99          | 0.86            | 0.3989       | 4      | 3          | 5         | 3  |

# How to conduct and report a Bayesian analysis – Reporting I

## 7. Estimation: Summarize the posterior

- Intervals and point estimates
- Ideally a plot of prior and posterior

## 8. Testing: Report the Bayes factor(s)

- Always with the subscript
- Posterior model probabilities with prior odds

## 9. Decision criteria

- What cutoff/decision rule are your conclusions based on?

## 10. Transparent reporting

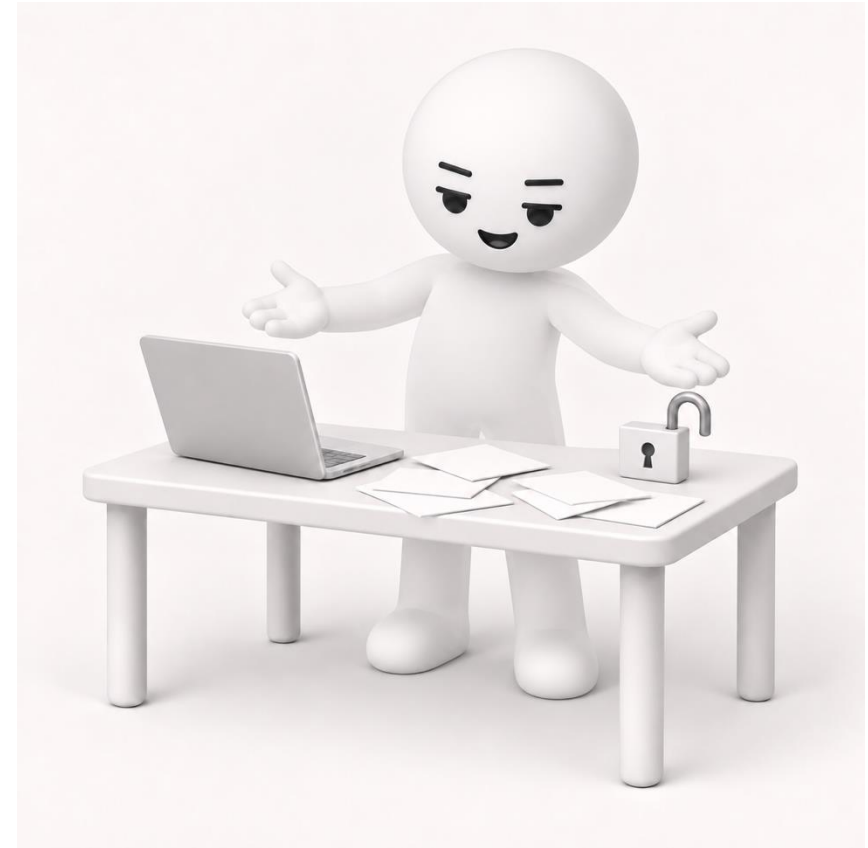
- Prior sensitivity analysis
- MCMC convergence results
- Preregistration deviations



# How to conduct and report a Bayesian analysis – Reporting II

## 11. Making it reproducible

- Software and version number(s)
- Seed (for MCMC procedures)
- Analysis scripts/files
- Anonymized data



# How to (not) conduct and report a Bayesian analysis – Examples

to our preregistration (i.e., without applying a correction to control the family-wise error rate).

This analysis yielded a significant interaction effect between headline veracity and treatment,  $\beta = 0.015$ , 95% CI = [0.0027, 0.027],  $F(3, 47490) = 4.52$ ,  $p = .017$ ; treatment-group effect size:  $d = -0.14$ , 95% CI = [-0.17, -0.12] (see Table S6 in the Supplemental Material). The effect size for the control group was directionally similar ( $d = -0.10$ , 95% CI = [-0.13, -0.078]), but sharing discernment was still 1.4 times higher in the treatment group than in the control group, an attenuation of about 50% compared with the effect sizes reported in the target study. Figure 1 shows the results in a bar graph.

Following our preregistration, we performed several additional analyses. First, we replicated the main analysis using high-dimensional fixed-effects linear regression (with STATA's *reghdfe* package), as we did in Stage 1 of data collection. Doing so also yielded a significant (uncorrected) effect for the interaction of headline veracity and treatment,  $\beta = 0.015$ , 95% CI = [0.0020, 0.028],  $F(3, 47490) = 4.52$ ,  $p = .026$  (see Table S6).

Second, we conducted independent-samples and Bayesian  $t$  tests to evaluate differences in sharing intentions between the treatment and control groups for discernment, as well as for each individual headline (Hypotheses 1 and 2). Doing so yielded a significant (noncorrected) effect for discernment (treatment:  $M = 0.26$ , control:  $M = 0.19$ , mean difference:  $-0.075$ , 95%

CI = [-0.15, -0.0019]),  $t(1581) = -2.013$ ,  $p = .044$ ,  $d = -0.10$ , 95% CI = [-0.20, -0.0025]. A Bayesian  $t$  test revealed a Bayes factor (BF) indicating that the data are approximately 1.7 times more likely to occur under the focal hypothesis than under the null hypothesis ( $BF_{10} = 1.705$ ,  $M = -0.061$ , 95% CI = [-0.17, 0.016], error percentage =  $7.148 \times 10^{-6}$ ), which is weak (or anecdotal) evidence in support of the focal hypothesis and against the null hypothesis that there would be no difference in truth discernment between the treatment and control groups (van Doorn et al., 2021; see Table S4 in the Supplemental Material).<sup>5</sup>

We found that—unlike in the target study—an independent-samples  $t$  test showed a significant difference for average willingness to share false headlines, in the sense that the treatment group was significantly less willing to share false headlines than the control group (treatment:  $M = 2.89$ , control:  $M = 3.04$ , mean difference:  $= 0.15$ , 95% CI = [0.011, 0.29]),  $t(1581) = 2.113$ ,  $p = .035$ ,  $d = 0.11$ , 95% CI = [0.0075, 0.21]. Similarly, a Bayesian  $t$  test gave weak support in favor of Hypothesis 1 ( $BF_{10} = 1.948$ ,  $M = 0.066$ , 95% CI = [-0.013, 0.170], error percentage = 0.0030). Also unlike the target study and against our Hypothesis 2, our analyses found no significant difference for real headlines (treatment:  $M = 3.15$ , control:  $M = 3.23$ , mean difference:  $0.075$ , 95% CI = [-0.057, 0.21]),  $t(1581) = 1.110$ ,  $p = .27$ ,  $d = 0.06$ , 95% CI = [-0.043, 0.15]. A Bayesian  $t$  test provided weak support against Hypothesis 2; treatment-group participants

In the analyses, we used the default priors provided by the *rstanarm*, *BEST*, and the *BayesFactor* packages. Specifically, in regression models we used the priors  $\mathcal{N}(0,10)$  for the intercept, and  $\mathcal{N}(0,2.5)$  for the coefficients. In the  $t$ -tests, we used the priors  $\mathcal{N}(\text{mean}(y), \text{sd}(y) * 1000)$  and  $\mathcal{U}(\text{sd}(y)/1000, \text{sd}(y) * 1000)$  for  $\mu$  and  $\sigma$ , and  $\mathcal{E}(1/29)$  for  $v$ , with  $v \geq 1$ . Finally, for correlations we used the prior *beta*(3,3).

As suggested by Makowski et al. (2019); we used the ROPE  $[-0.1SD_y, 0.1SD_y]$  for testing Hypothesis 1, Hypothesis 2, Hypothesis 3, and the ROPE  $[-.05, .05]$  for testing Hypothesis 4. When reporting parameters, we report the median and the 95% HDI of the posterior distribution, as well as the proportion of the posterior distribution that lies within the ROPE (pROPE; note that we interpret the evidence to be conclusive if this value is smaller than .025).



# How to (not) conduct and report a Bayesian analysis – Examples

2018), we also conducted an analysis following a Bayesian approach using JASP Version 0.9.2 (JASP Team, 2019). The Bayes factor (BF) compares the probability of the data under one model with the data under another model and indicates which hypothesis is more likely, given the data (Quintana & Williams, 2018). Following Jarosz and Wiley's (2014) guidelines, we considered a Bayes value of 3.0 or greater as providing significant evidence in support of the null hypothesis and considered a Bayes value of less than 0.33 as supporting the alternative hypothesis. The Bayesian analysis yielded a  $BF_{01}$  of 6.65, thereby providing support for the null hypothesis.

We coded participants' reports of their own emotions as either matching or not matching the primed emotion. The proportion of participants who chose a photograph that matched the primed emotion did not differ by emotion condition,  $\chi^2(1, N = 98) = 0.20, p = .658$ . That is, 44% of participants in the pride condition and 40% of participants in the joy condition chose the matched emotion. Bayesian analysis yielded a  $BF_{01}$  of 3.72, providing support for the null hypothesis. As reported previously, the distribution of children's reported emotions was similar across conditions; thus, we can conclude that children's current emotions were not dependent on the primed emotion.

Pick, 1993; Thibault et al., 2013; Tlauka et al., 2007; Wilson et al., 2004). To test whether there was evidence in favor of the null hypothesis (i.e., that spatial memory was similar for the two planes), we examined the data by conducting a Bayesian  $t$  test, and therefore obtained Bayes factors using the Bayesian information criteria. The estimated Bayes factors were  $BF_{01} = 3.81$  for error and  $BF_{01} = 3.42$  for latency, indicating positive evidence (according to Raftery, 1995) in favor of the null hypothesis. If spatial relations have been learned from a single perspective, memory for horizontal and memory for vertical locations seem to yield similar performance.

---

<sup>1</sup> We also performed supplementary hierarchical Bayesian analyses for Experiments 1 and 2 with TreeBUGS (Heck et al., 2018) that take individual variability into account, which confirmed the results based on the aggregated data.

# How to (not) conduct and report a Bayesian analysis – Examples

ANOVA. The main effects and interactions between factors were analyzed, and the results are presented below. The interactions were indeed tested in the raw error rates and RTs. However, in the subsequent decomposition, we converted the error rate and RT data into magnitudes to facilitate the interpretation and visualization of the interaction effects. This conversion was solely for the purpose of presenting the results in a more accessible format. Bayesian ANOVAs were also performed, using default priors within R. Bayes Factors (BFs) provide an estimate of the strength of evidence for the data under the null and alternative hypotheses. For ANOVA,  $BF_{incl}$  are used to measure the strength of evidence for the inclusion of each factor and interaction in the model, including  $BF_{01}$  to assess evidence for the null hypothesis. For  $t$  tests,  $BF_{10}$  are reported, indicating evidence for the presence of an effect, with  $BF_{01}$  assessing evidence for the absence of an effect.  $BF < 1$  indicates support for the null hypothesis, while  $BF > 1$  supports the alternative hypothesis. We adopt the commonly used classification scheme, where  $BF$  1–3 equates to anecdotal evidence,  $BF$  3–10 as moderate evidence,  $BF$  10–100 as strong evidence, and  $BF > 100$  as very strong evidence (Lee & Wagenmakers, 2014; van Doorn et al., 2021).

**Error rates** A four-factor repeated-measures ANOVA showed that a significant main effect of current congruency,  $F(1, 29) = 12.03, p = .002, \eta_p^2 = 0.29, BF > 100$ . The error rates were higher in the incongruent condition (6.0%) than in the congruent condition (3.9%). The main effect of the previous currency was also significant,  $F(1, 29) = 22.35, p < .001, \eta_p^2 = 0.44, BF = 2.40$ . The previous congruent trials (5.4%) were higher error rate than the previous incongruent trials (4.5%). The main effect of the previous self-referential information was not significant,  $F(1, 29) = 4.00, p = .055, \eta_p^2 = 0.12, BF = 0.50 (BF_{01} = 2.00)$ . The main effect of the current self-referential information was not significant,  $F(1, 29) < 1, BF = 0.08$ .

The interaction between the previous congruency and the current congruency was not significant,  $F(1, 29) = 3.78, p = .062, \eta_p^2 = 0.11, BF = 0.98 (BF_{01} = 1.02)$ . The interaction between the previous congruency and the previous self-referential information was not significant,  $F(1, 29) = 4.17, p = .050, \eta_p^2 = 0.13, BF = 0.63 (BF_{01} = 1.59)$ . The interaction between the current congruency and the previous self-referential information was not significant,  $F(1, 29) = 3.04, p = .092, \eta_p^2 = 0.09, BF = 0.40 (BF_{01} = 2.50)$ . The interaction between the previous congruency and the current self-referential information was not significant,  $F(1, 29) < 1, BF = 0.16$ . The interaction between the current congruency and the current self-referential information was not significant,  $F(1, 29) = 1.77, p = .062, \eta_p^2 = 0.19, BF = 0.19 (BF_{01} = 5.26)$ . The interaction between the previous self-referential information and the current self-referential information was

# How to (not) conduct and report a Bayesian analysis – Examples

Models were fitted using the R package qgam (Fasiolo et al., 2021a). Since initial model fits indicated problems due to complete separation,<sup>1</sup> we followed Donnelly and Verkuilen (2017) in adding a small smoothing constant, which we set to 0.1, to our models. Significance of the terms in the models was established by predicting each model's linear-predictor matrix onto a grid of time points ranging from 0 to 1,000 ms in both quiet and noise conditions. Using the procedure in Wood (2017: 293–294), 95% Bayesian credible intervals (henceforth: CIs) were computed for the predicted log odds ratios in both quiet and noise conditions, as well as for the difference between these. For the target-absent models, additional differences between the two competitors were computed in the same way. Effects were considered to be significantly above/below chance at time points where their CIs excluded zero.

of incentives, we used JASP in Version 0.15 (JASP Team, 2021). A Bayesian repeated-measures ANOVA showed a Bayes factor of  $BF_{01} = 20.632$ , commonly seen as strong evidence for the  $H_0$  (Jarosz & Wiley, 2014). As JASP lacks functionality for custom contrasts, we used R (R Core Team, 2020) with the packages rstanarm (Goodrich et al., 2020) and bayestestR (Makowski et al., 2019) for the exploratory –2, 1, 1 contrast. This analysis showed a Bayes factor of  $BF_{01} = 6.410$ , commonly seen as positive or substantial evidence for the  $H_0$  (Jarosz & Wiley, 2014). The JASP file containing the analysis and the relevant R code are accessible from the OSF project linked above.



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1. Bayesian multi-model inference
2. Mixed-effects regression in R with *brms*
3. How to conduct and report a Bayesian analysis
- 4. Bring your own data**
5. Beyond the workshop
6. Conclusions

# Bring your own data

-> Open guide.md in the GitHub repository under bring-your-own-data and follow the instructions there

# Contents – Day 2

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# Beyond the workshop

- Bayesian meta-analysis in JASP
- BFpack in JASP
- Bayesian network analysis in JASP
- BFDA in R
- Blavaan in R
- Stan in R
- Bridgesampling in R
- Spike and slab priors

# Beyond the workshop – Bayesian meta-analysis in JASP

## Purpose

- Synthesizing results across studies

## Why Bayesian:

- Evidence for the null: Quantify how likely it is that the effect is absent across studies
- Publication bias modelled jointly with the effect size estimation
- Model averaging across meta-analytic models

## Resource:

- <https://jasp-stats.org/2020/05/12/how-to-conduct-a-bayesian-model-averaged-meta-analysis-in-jasp/>
- <https://jasp-stats.org/2020/10/22/accounting-for-publication-bias-with-robust-bayesian-meta-analysis-in-jasp/>



## Bayesian

Meta-Analysis

Meta-Analysis (Deprecated)

Binomial Meta-Analysis

Penalized Meta-Analysis

Prediction Model Performance

Robust Bayesian Meta-Analysis

# Beyond the workshop – BFpack in JASP

Purpose:

- Test hypotheses involving constraints jointly
- E.g.:
  - $H_1$ : Controls = Treatment1 < Treatment2
  - $H_2$ : Controls < Treatment1 = Treatment2
  - $H_3$ : Controls = Treatment1 = Treatment2
  - $H_4$ : Complement

Why Bayesian:

- Bayes factors and posterior model probabilities readily available

Resource:

- Mulder, J., & Pfadt, J. M. (2026). Going in the right direction: A tutorial to directional hypothesis testing using the BFpack module in JASP. *Advances in Methods and Practices in Psychological Science*, 9(3).  
<https://doi.org/10.1177/25152459261463155>

|                                                                                       |                                           |
|---------------------------------------------------------------------------------------|-------------------------------------------|
|    | <b>t-Tests</b>                            |
|                                                                                       | Independent Samples t-Test                |
|                                                                                       | Paired Samples t-Test                     |
|                                                                                       | One Sample t-Test                         |
|                                                                                       | Multivariate t-Test                       |
|    | <b>ANOVA</b>                              |
|                                                                                       | (M)AN(C)OVA                               |
|    | <b>Regression</b>                         |
|                                                                                       | Univariate/Multivariate Linear Regression |
|                                                                                       | (Ordered) Logistic Regression             |
|                                                                                       | Correlation                               |
|  | <b>Variances</b>                          |
|                                                                                       | Variances                                 |

# Beyond the workshop – Bayesian networks in JASP

Purpose:

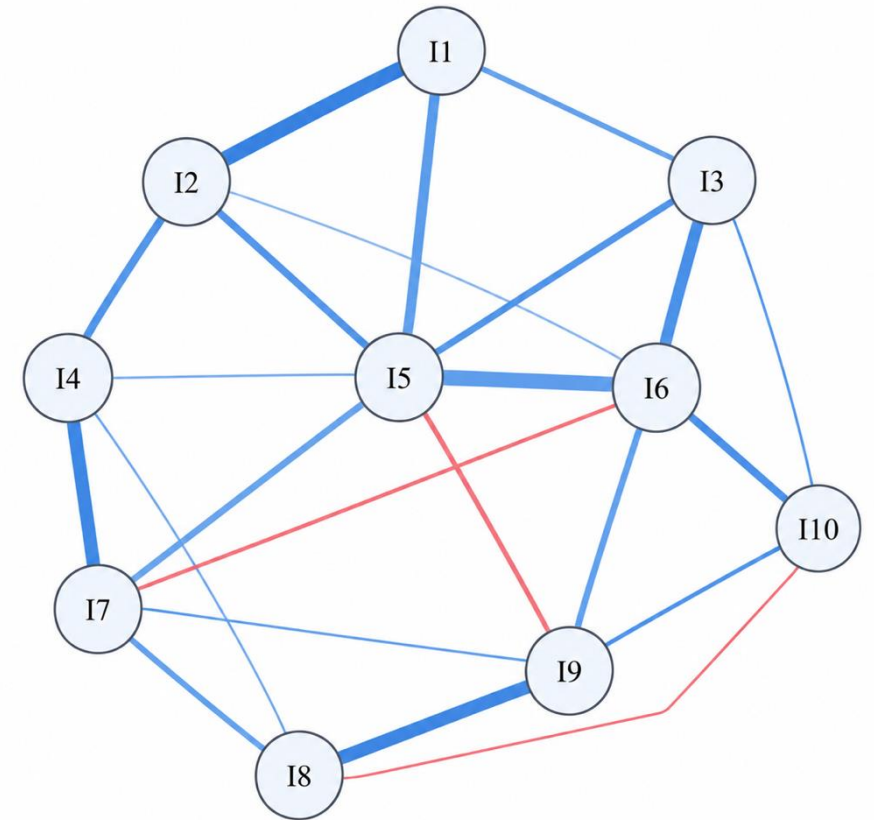
- Model psychological constructs as systems of interacting variables; also used in neuroscience and finance

Why Bayesian:

- Structural uncertainty: A posterior over network structure, not a single choice
- Evidence for edge inclusion or exclusion

Resource:

- Huth, K. B. S., de Ron, J., Goudriaan, A. E., Luigjes, J., Mohammadi, R., van Holst, R. J., Wagenmakers, E.-J., & Marsman, M. (2023). Bayesian analysis of cross-sectional networks: A tutorial in R and JASP. *Advances in Methods and Practices in Psychological Science*, 6(4).  
<https://doi.org/10.1177/25152459231193334>





# Beyond the workshop – BFDA (Bayes factor design analysis) in R

Purpose:

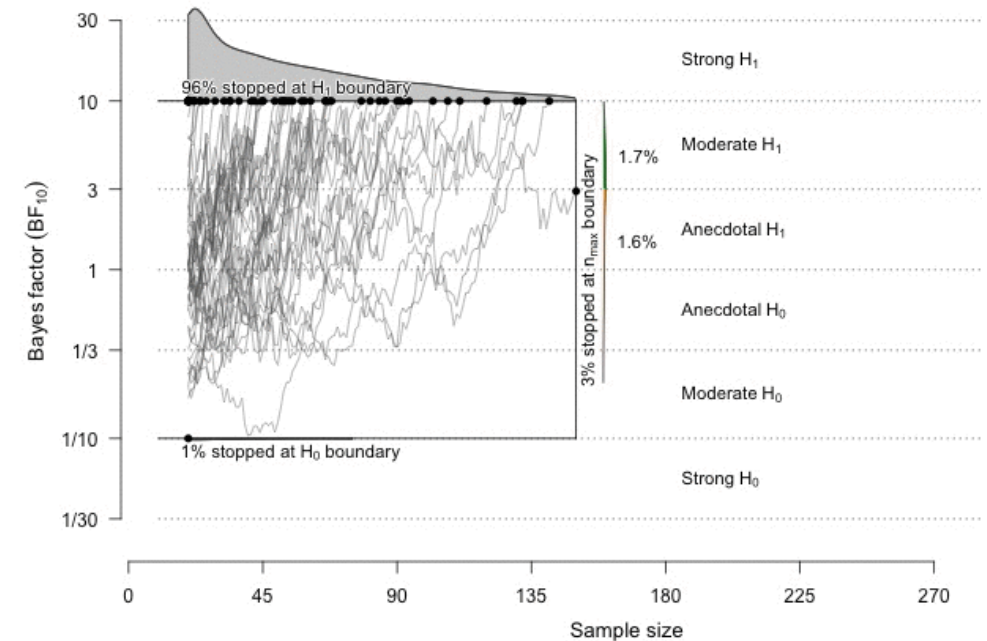
- Sample size planning and data-collection strategy

Why Bayesian:

- Given an assumed effect size, how much data is needed to reach a chosen strength of evidence?

Resource:

- Stefan, A. M., Gronau, Q. F., Schönbrodt, F. D., & Wagenmakers, E.-J. (2019). A tutorial on Bayes Factor Design Analysis using an informed prior. *Behavior Research Methods*, 51(3), 1042–1058.  
<https://doi.org/10.3758/s13428-018-01189-8>



# Beyond the workshop – blavaan in R

Purpose:

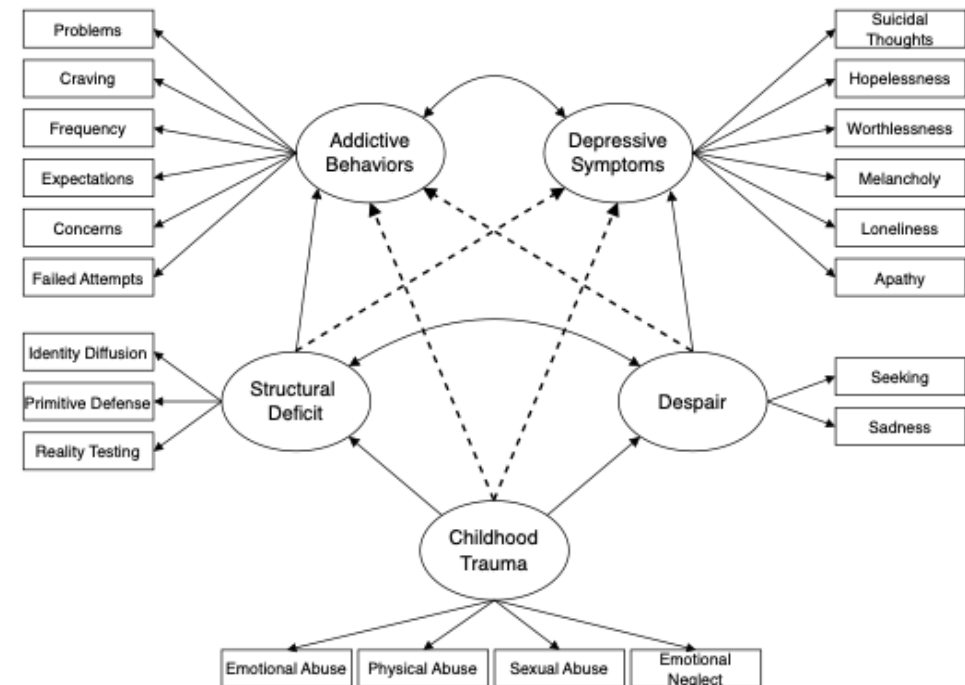
- Factor analysis and structural equation modeling
- Estimating and modeling latent variables from observed data

Why Bayesian?

- Posterior distributions
- Bayes factors and model averaging

Resource:

- <https://blavaan.org/articles/start.html>
- Pfadt, J. M., Merkle, E. C., & Wagenmakers, E.-J. (2026). *Bayes factors for structural equation models with bridge sampling and blavaan*. PsyArXiv. [https://doi.org/10.31234/osf.io/pt2bc\\_v1](https://doi.org/10.31234/osf.io/pt2bc_v1)



# Beyond the workshop – Stan in R

- A probabilistic programming language in the background of *brms* or *blavaan*
- Does all the MCMC sampling for you
- In R via *rstan* or *cmdstanr*
- Obtain the posterior distribution of any model for which you can formulate the priors and the likelihood
- With LLMs writing Stan code is relatively easy, but be cautious

Resource:

- <https://mc-stan.org/learn-stan/tutorials.html>

```
bernoulli.stan

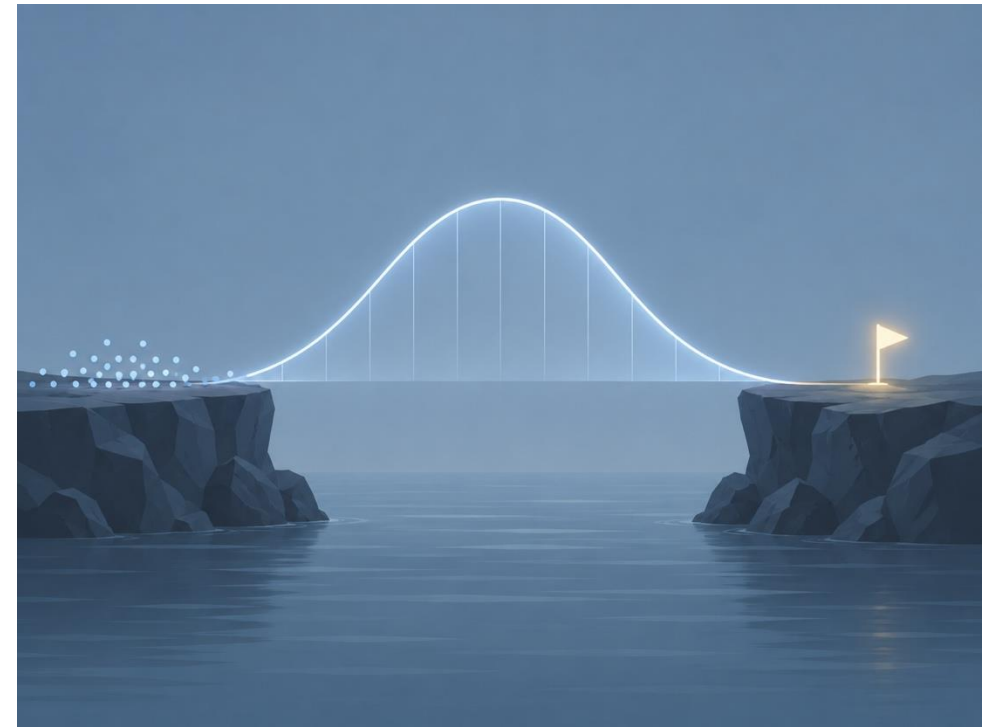
data {
  int<lower=0> N;
  array[N] int<lower=0, upper=1> y;
}
parameters {
  real<lower=0, upper=1> theta;
}
model {
  // uniform prior on interval 0,1
  theta ~ beta(1, 1);
  y ~ bernoulli(theta);
}
```

# Beyond the workshop – Bridge sampling in R

- Estimating marginal likelihood using the posterior samples plus samples from a reference distribution (the bridge)
- The R package accepts output from, e.g., *Stan* fits, *brms* fits, or *blavaan* fits

Resource:

- Gronau, Q. F., Singmann, H., & Wagenmakers, E.-J. (2020). bridgesampling: An R package for estimating normalizing constants. *Journal of Statistical Software*, 92(10), 1–29.  
<https://doi.org/10.18637/jss.v092.i10>

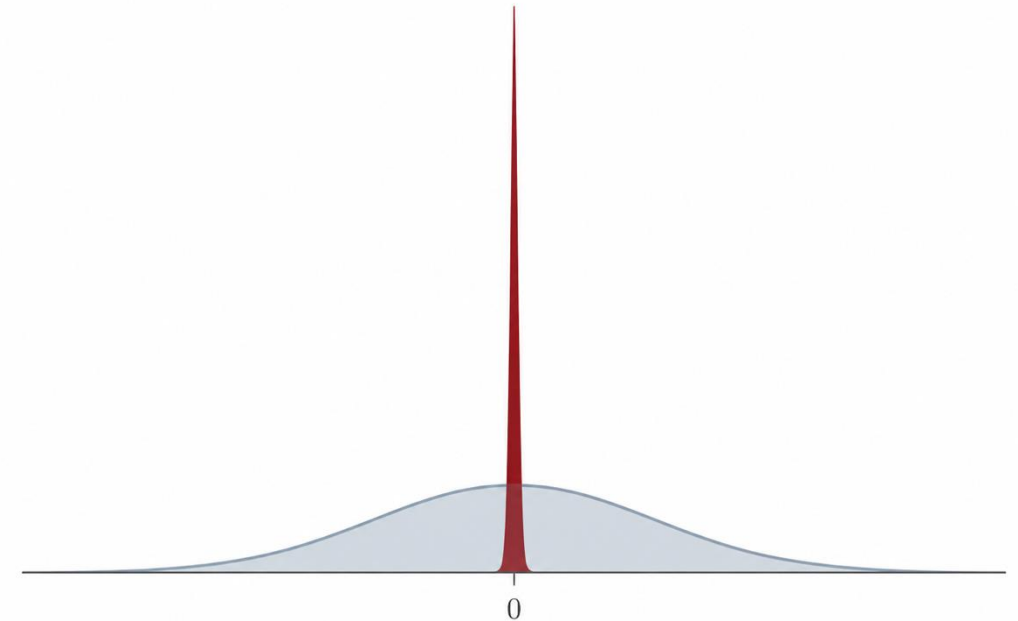


# Beyond the workshop – Spike-and-slab priors

- Decide which predictors belong in a model
- The decision mechanism is already build in the prior construction
- Each coefficient gets a priors that is a mixture of two components:
  - A spike: narrow distribution concentrated at zero
  - A slab: wide distribution
- Posterior mass concentrates in two regions
- If more posterior mass concentrates around the spike (0), it is more likely the coefficient is excluded, if it concentrates away from the spike it is more likely the predictor is included

Resource:

- <https://jasp-stats.org/2024/04/03/bayesian-state-space-models-in-jasp/>
- Van Erp, S., Oberski, D. L., & Mulder, J. (2019). Shrinkage priors for Bayesian penalized regression. *Journal of Mathematical Psychology*, 89, 31–50.  
<https://doi.org/10.1016/j.jmp.2018.12.004>



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# Ongoing discussions in the Bayesian world

- Estimation vs Testing
- Posterior predictive performance vs Bayes factors
- The prior is the devil



# Further reading

- MCMC: Van Ravenzwaaij, D., Cassey, P., & Brown, S. D. (2018). A simple introduction to Markov Chain Monte-Carlo sampling. *Psychonomic Bulletin & Review*, 25(1), 143–154. <https://doi.org/10.3758/s13423-016-1015-8>
- *brms*:
  - <https://paulbuerkner.com/brms/>
  - Paul-Christian Bürkner (2017). brms: An R Package for Bayesian Multilevel Models Using Stan. *Journal of Statistical Software*, 80(1), 1-28. doi:10.18637/jss.v080.i01z
- How to conduct and report a Bayesian analysis:
  - Kruschke, J. K. (2021). Bayesian analysis reporting guidelines. *Nature Human Behaviour*, 5, 1282–1291. <https://doi.org/10.1038/s41562-021-01177-7>
  - Van Doorn, J., Van den Bergh, D., Böhm, U., Dablander, F., Derks, K., Draws, T., Etz, A., Evans, N. J., Gronau, Q. F., Haaf, J. M., Hinne, M., Kucharský, Š., Ly, A., Marsman, M., Matzke, D., Komarlu Narendra Gupta, A. R., Sarafoglou, A., Stefan, A., Voelkel, J. G., & Wagenmakers, E.-J. (2021). The JASP guidelines for conducting and reporting a Bayesian analysis. *Psychonomic Bulletin & Review*, 28, 813–826. <https://doi.org/10.3758/s13423-020-01798-5>



# Discussion & Feedback

- Do you see yourself applying Bayesian statistics in your own research?
- If so, how?
- If not, why?
- What do you struggle with?